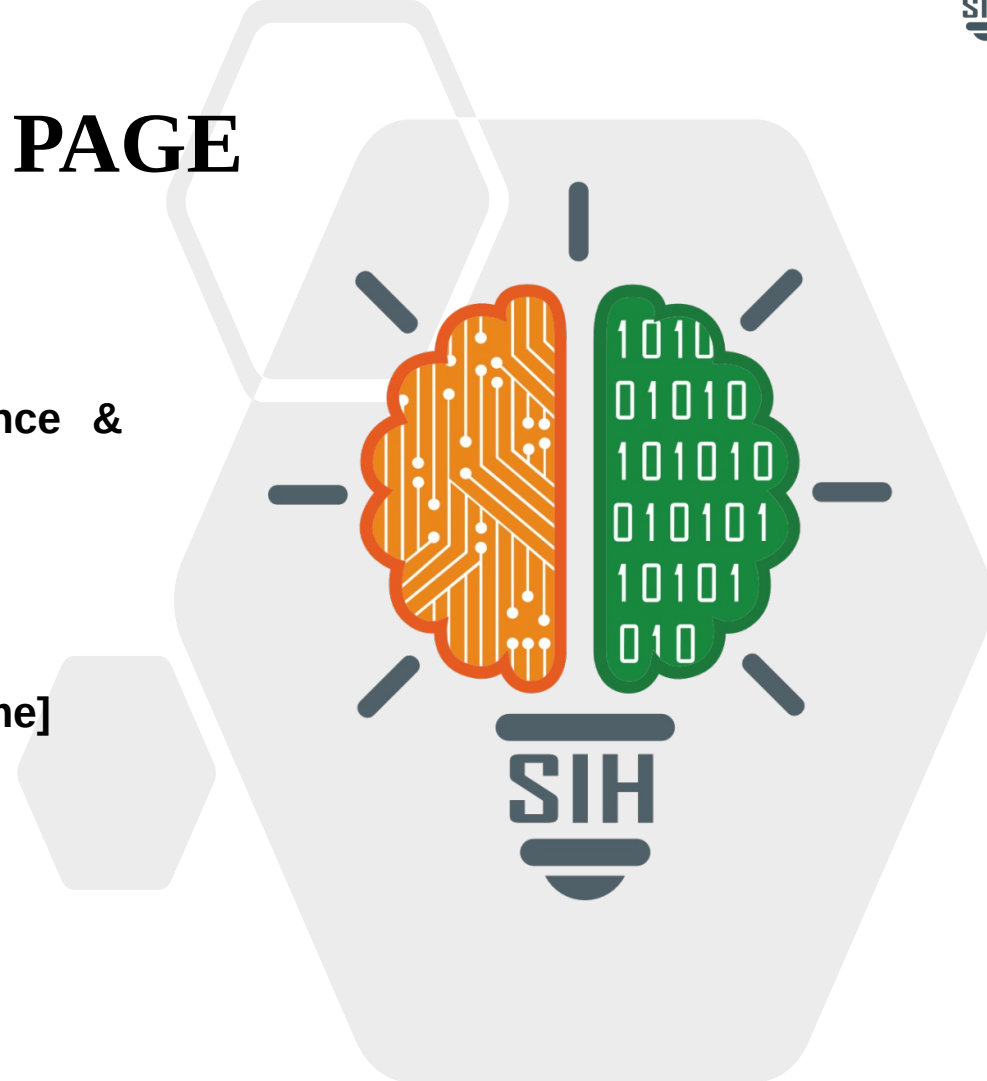




TITLE PAGE

- Problem Statement ID – [Add PS ID from portal]
- Problem Statement Title- Proof-Based Attendance & Fraud Detection System
- Theme- Smart Education
- PS Category- Hardware
- Team ID- [Add Team ID]
- Team Name (Registered on portal)- [Add Team Name]



❖ PRAMAAN – Proof-Based Attendance & Fraud Detection System

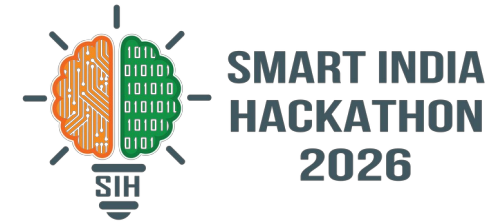
- Every classroom gets a small Bluetooth node that listens for a signal from wearable bands worn by students and teachers. The band only starts broadcasting once it reads a real fingerprint, and that signal hops room to room until it reaches one gateway connected to the internet
- This is aimed straight at the two things actually costing schools money: inflated student counts used to claim extra scholarship funds, and staff marking themselves present and then leaving. Nobody can quietly edit a record afterwards either — only an admin can, and even then it gets logged
- As far as we could find, nobody has put mesh networking, on-the-band fingerprint checks, and always-on presence tracking together for both students and staff in one system — and because it's a mesh, you don't need Wi-Fi wired into every classroom

- We're building the mesh nodes and gateway on ESP32 boards talking to each other over ESP-NOW, with a fingerprint sensor on each wearable band. The backend runs on Node.js and Express, SQLite for now (Postgres once this scales), a React dashboard, and a Flutter app for phones
- Here's the flow: fingerprint matches on the band, the band sends out a verified signal, the nearest room node picks it up and passes it along the mesh, it reaches the gateway, and the backend logs



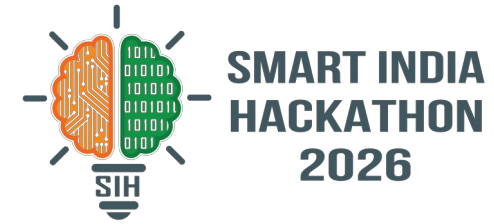
You only need real internet at one point in the building — if a node goes down, the mesh just routes around it

FEASIBILITY AND VIABILITY



- Nothing here is exotic — ESP32 boards, Bluetooth, basic fingerprint sensors are all cheap and easy to source. We'd also roll it out in stages: staff bands first, then simple tags for students, then fingerprint bands for students once there's budget for it
- The risks we're most aware of: fingerprint sensors don't read young kids' fingers reliably, someone could try to record and replay a Bluetooth signal, biometric data on minors needs careful handling, and rural schools often don't have anyone around who can fix broken hardware
- Strategies: we'd keep fingerprint bands to staff and older students, use rotating codes instead of one fixed signal so nothing can be recorded and replayed, never let raw fingerprint data leave the device, and lean on the IT support states already have in place for other school programs

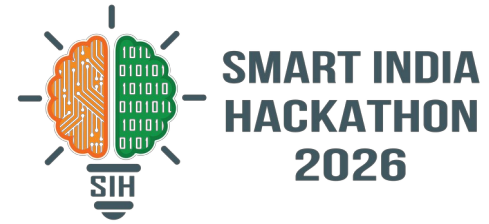
IMPACT AND BENEFITS



- Students end up in a system where scholarships actually reach the people who show up. Teachers stop getting blamed for someone else's manual mistakes. And district officers can finally see real numbers instead of waiting months for an audit to catch a problem
- Socially, it means scholarships reach the students they were actually meant for. Financially, it protects welfare funding from being claimed on students who don't really attend. And it replaces a paper trail anyone could alter with one that can't be quietly changed

[Team
Name]

RESEARCH AND REFERENCES



- We looked at the DoPT's AEBAS mandate from 2023, attendance circulars from Punjab, Himachal Pradesh, Haryana and Karnataka, the Digital Personal Data Protection Act of 2023, and reports on how BLE mesh and fingerprint attendance are already used in other industries